

COAL Lab-2

Screenshots of Watch

Q-1:

Operations on nx before and after:

Watch 1

Search (Ctrl+E)  Search Depth: 3

Name	Value	Type
 ax	2	unsigned short
 bx	10	unsigned short
 cx	3	unsigned short
 dx	5	unsigned short
 al	2 '\x2'	unsigned char
 ah	0 '\0'	unsigned char
 bl	10 '\n'	unsigned char
 bh	0 '\0'	unsigned char
 cl	3 '\x3'	unsigned char
 ch	0 '\0'	unsigned char
 dl	5 '\x5'	unsigned char
 dh	0 '\0'	unsigned char
Add item to watch		

Watch 1

Search (Ctrl+E)  Search Depth: 3

Name	Value	Type
 ax	5	unsigned short
 bx	5	unsigned short
 cx	3	unsigned short
 dx	5	unsigned short
 al	5 '\x5'	unsigned char
 ah	0 '\0'	unsigned char
 bl	5 '\x5'	unsigned char
 bh	0 '\0'	unsigned char
 cl	3 '\x3'	unsigned char
 ch	0 '\0'	unsigned char
 dl	5 '\x5'	unsigned char
 dh	0 '\0'	unsigned char
Add item to watch		

Operations on nl, nh before and after:

Watch 1

Search (Ctrl+E) Search Depth: 3

Name	Value	Type
ax	3598	unsigned short
bx	514	unsigned short
cx	1287	unsigned short
dx	1030	unsigned short
al	14 '\xe'	unsigned char
ah	14 '\xe'	unsigned char
bl	2 '\x2'	unsigned char
bh	2 '\x2'	unsigned char
cl	7 '\a'	unsigned char
ch	5 '\x5'	unsigned char
dl	6 '\x6'	unsigned char
dh	4 '\x4'	unsigned char
Add item to watch		

Watch 1

Search (Ctrl+E) Search Depth: 3

Name	Value	Type
ax	1801	unsigned short
bx	1544	unsigned short
cx	1287	unsigned short
dx	1030	unsigned short
al	9 '\t'	unsigned char
ah	7 '\a'	unsigned char
bl	8 '\b'	unsigned char
bh	6 '\x6'	unsigned char
cl	7 '\a'	unsigned char
ch	5 '\x5'	unsigned char
dl	6 '\x6'	unsigned char
dh	4 '\x4'	unsigned char
Add item to watch		

Q-2:

Operations on nx before and after:

Watch 1

Search (Ctrl+E)  Search Depth: 3

Name	Value	Type
 ax	2	unsigned short
 bx	10	unsigned short
 cx	5	unsigned short
 dx	5	unsigned short
 al	2 '\x2'	unsigned char
 ah	0 '\0'	unsigned char
 bl	10 '\n'	unsigned char
 bh	0 '\0'	unsigned char
 cl	5 '\x5'	unsigned char
 ch	0 '\0'	unsigned char
 dl	5 '\x5'	unsigned char
 dh	0 '\0'	unsigned char
Add item to watch		

Watch 1

Search (Ctrl+E)  Search Depth: 3

Name	Value	Type
 ax	7	unsigned short
 bx	5	unsigned short
 cx	5	unsigned short
 dx	5	unsigned short
 al	7 '\a'	unsigned char
 ah	0 '\0'	unsigned char
 bl	5 '\x5'	unsigned char
 bh	0 '\0'	unsigned char
 cl	5 '\x5'	unsigned char
 ch	0 '\0'	unsigned char
 dl	5 '\x5'	unsigned char
 dh	0 '\0'	unsigned char
Add item to watch		

Operations on nl, nh before and after:

Watch 1

Search (Ctrl+E)  Search Depth: 3

Name	Value	Type
 ax	1801	unsigned short
 bx	1544	unsigned short
 cx	1285	unsigned short
 dx	1285	unsigned short
 al	9 '\t'	unsigned char
 ah	7 '\a'	unsigned char
 bl	8 '\b'	unsigned char
 bh	6 '\x6'	unsigned char
 cl	5 '\x5'	unsigned char
 ch	5 '\x5'	unsigned char
 dl	5 '\x5'	unsigned char
 dh	5 '\x5'	unsigned char
Add item to watch		

Watch 1

Search (Ctrl+E)  Search Depth: 3

Name	Value	Type
 ax	3086	unsigned short
 bx	259	unsigned short
 cx	1285	unsigned short
 dx	1285	unsigned short
 al	14 '\xe'	unsigned char
 ah	12 '\f'	unsigned char
 bl	3 '\x3'	unsigned char
 bh	1 '\x1'	unsigned char
 cl	5 '\x5'	unsigned char
 ch	5 '\x5'	unsigned char
 dl	5 '\x5'	unsigned char
 dh	5 '\x5'	unsigned char
Add item to watch		

Q-3:

Operations before and after:

Watch 1			▼	📌	✕
Search (Ctrl+E)			🔍 ▼	←	→
			Search Depth: 3 ▼		
Name	Value	Type			
 ax	1092	unsigned...			
 bx	5	unsigned...			
 al	68 'D'	unsigned...			
 ah	4 '\x4'	unsigned...			
 bl	5 '\x5'	unsigned...			
 bh	0 '\0'	unsigned...			
Add item to watch					

Watch 1			▼	📌	✕
Search (Ctrl+E)			🔍 ▼	←	→
			Search Depth: 3 ▼		
Name	Value	Type			
 ax	1097	unsigned...			
 bx	5	unsigned...			
 al	73 'I'	unsigned...			
 ah	4 '\x4'	unsigned...			
 bl	5 '\x5'	unsigned...			
 bh	0 '\0'	unsigned...			
Add item to watch					

Q-4:

Operations before and after:

 cx	523	unsigned...
 dx	617	unsigned...
 cl	11 '\v'	unsigned...
 ch	2 '\x2'	unsigned...
 dl	105 'i'	unsigned...
 dh	2 '\x2'	unsigned...
Add item to watch		

 cx	1140	unsigned...
 dx	617	unsigned...
 cl	116 't'	unsigned...
 ch	4 '\x4'	unsigned...
 dl	105 'i'	unsigned...
 dh	2 '\x2'	unsigned...
Add item to watch		

Q-5:

Operations before and after:

 cx	2024	unsigned...
 dx	2005	unsigned...
 cl	232 'è'	unsigned...
 ch	7 '\a'	unsigned...
 dl	213 'Ö'	unsigned...
 dh	7 '\a'	unsigned...
Add item to watch		

 cx	19	unsigned...
 dx	2005	unsigned...
 cl	19 '\x13'	unsigned...
 ch	0 '\0'	unsigned...
 dl	213 'Ö'	unsigned...
 dh	7 '\a'	unsigned...
Add item to watch		

Q-6:

Operations before and after:

 cx	364	unsigned...
 dx	436	unsigned...
 cl	108 'l'	unsigned...
 ch	1 '\x1'	unsigned...
 dl	180 ''	unsigned...
 dh	1 '\x1'	unsigned...
Add item to watch		

 cx	800	unsigned...
 dx	436	unsigned...
 cl	32 ''	unsigned...
 ch	3 '\x3'	unsigned...
 dl	180 ''	unsigned...
 dh	1 '\x1'	unsigned...
Add item to watch		

Q-7:

Operations before and after:

Watch 1			▼	🔍	×
Search (Ctrl+E)			🔍	← →	Search Depth: 3 ▼
Name	Value	Type			
 cx	2052	unsigned...			
 dx	2730	unsigned...			
 cl	4 '\x4'	unsigned...			
 ch	8 '\b'	unsigned...			
 dl	170 'a'	unsigned...			
 dh	10 '\n'	unsigned...			
Add item to watch					

Watch 1			▼	🔍	×
Search (Ctrl+E)			🔍	← →	Search Depth: 3 ▼
Name	Value	Type			
 cx	4782	unsigned...			
 dx	2730	unsigned...			
 cl	174 '®'	unsigned...			
 ch	18 '\x12'	unsigned...			
 dl	170 'a'	unsigned...			
 dh	10 '\n'	unsigned...			
Add item to watch					

Q-8:

	Code	Description	Project	File	Line	De
✖		'23i-0523_Moiz Ansari_Q6.asm' in 'Lab-2 Project File.exe' was not compiled with Edit and Continue enabled. Ensure that the file is compiled with the Program Database for Edit and Continue (/ZI) option at Project Properties > C/C++ > Debug Information Format.		Hot Reload		
✖	A2070	invalid instruction operands	Lab-2 Project File	23i-0523_Moiz Ansari_Q8....	15	
✖	MSB3721	The command "ml.exe /c /nologo /Zi /Fo"Debug\23i-0523_Moiz Ansari_Q8.obj" /W3 /errorReport:prompt /Ta"23i-0523_Moiz Ansari_Q8.asm"" exited with code 1.	Lab-2 Project File	masm.targets	70	

Reason:

The two registers will not add the values because ax is of 16 bits, while bl is of 8 bits. Since both registers should be of the same size to add so they won't add up.

Q-9:

Operations before and after:

 cx	186	unsigned...
 dx	618	unsigned...
 cl	186 'o'	unsigned...
 ch	0 '\0'	unsigned...
 dl	106 'j'	unsigned...
 dh	2 '\x2'	unsigned...
Add item to watch		

 cx	804	unsigned...
 dx	618	unsigned...
 cl	36 '\$'	unsigned...
 ch	3 '\x3'	unsigned...
 dl	106 'j'	unsigned...
 dh	2 '\x2'	unsigned...
Add item to watch		

Reason:

They would add together because their sizes are same, just their representation is different when inputting them. Inside the registers they are stored in the same format.

Q-10:

Operations before and after:

 cx	186	unsigned...
 dx	618	unsigned...
 cl	186 'o'	unsigned...
 ch	0 '\0'	unsigned...
 dl	106 'j'	unsigned...
 dh	2 '\x2'	unsigned...
Add item to watch		

 cx	804	unsigned...
 dx	618	unsigned...
 cl	36 '\$'	unsigned...
 ch	3 '\x3'	unsigned...
 dl	106 'j'	unsigned...
 dh	2 '\x2'	unsigned...
Add item to watch		

Q-11:

Operations before and after:

 cx	186	unsigned...
 dx	618	unsigned...
 cl	186 'o'	unsigned...
 ch	0 '\0'	unsigned...
 dl	106 'j'	unsigned...
 dh	2 '\x2'	unsigned...
Add item to watch		

 cx	804	unsigned...
 dx	618	unsigned...
 cl	36 '\$'	unsigned...
 ch	3 '\x3'	unsigned...
 dl	106 'j'	unsigned...
 dh	2 '\x2'	unsigned...
Add item to watch		

Q-12:

Watch 1		
Search (Ctrl+E) 🔍		
Search Depth: 3		
Name	Value	Type
al	9 '\t'	unsigned...
ah	6 '\x6'	unsigned...
Result	15 '\xf'	unsigned...
Add item to watch		

Q-13:

Watch 1		
Search (Ctrl+E) 🔍		
Search Depth: 3		
Name	Value	Type
al	9 '\t'	unsigned...
ah	6 '\x6'	unsigned...
Result	3 '\x3'	unsigned...
Add item to watch		

Q-14:

Watch 1		
Search (Ctrl+E) 🔍		
Search Depth: 3		
Name	Value	Type
type_of_str	1 '\x1'	unsigned...
L	7 '\a'	unsigned...
S	7 '\a'	unsigned...
myword	66 'B'	unsigned...
Add item to watch		